CL1002 P***rogramming Fundamentals Lab***	Lab 01 Introduction to Programming Fundamentals

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Fall 2024

AIMS AND OBJECTIVES

The aim of this lab is to equip students with the foundational knowledge and skills necessary to understand and apply fundamental programming concepts. This includes developing problem-solving abilities using real-world examples and familiarizing students with various tools and methodologies such as PAC (Problem Analysis Chart), IPO (Input-Process-Output) Charts and Flowcharts.

INTRODUCTION

In this lab we will be covering these topics

1. Problem solving in the context of programming fundamentals.
2. Introduction to Problem Analysis Charts.
3. Introduction to Input Process Output Table.
4. Introduction to flowcharts.

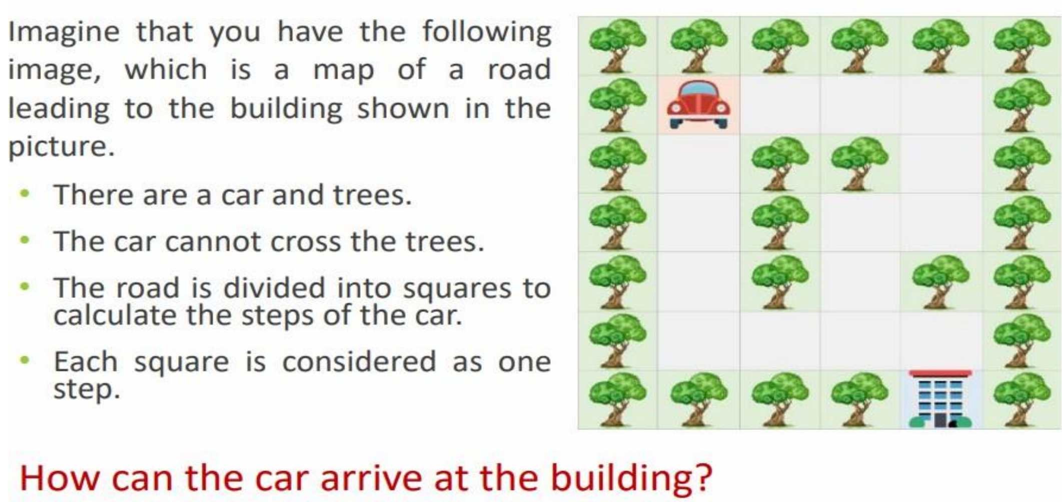
PROBLEM SOLVING

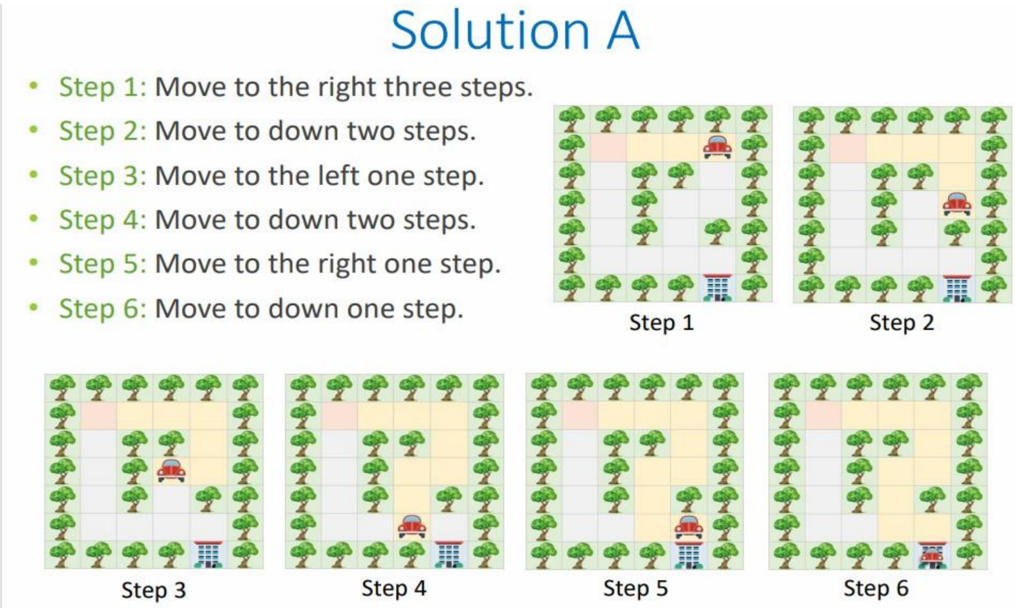
What is Problem solving.

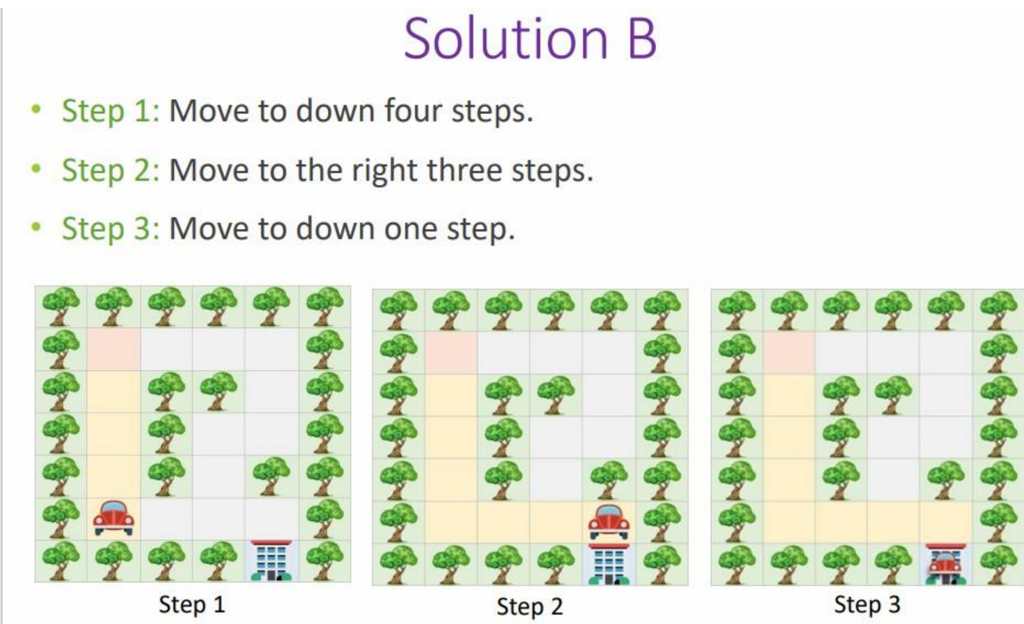
• **Definition**: Problem-solving in programming is identifying a problem, planning a solution, and executing that solution to achieve the desired result.

• **Importance**: It's fundamental in programming because it helps create efficient, effective, and scalable solutions.

EXAMPLE







As we can see **Solution A** and **Solution B** are both correct solutions to the same problem, but there are differences in the complexity and efficiency of the solutions.

The cost of Solution A is 10 steps while Solution B is 8, so we consider Solution B a better solution based on the number of steps.

Reducing the number of steps in the previous example means reducing the fuel needed by the vehicle and speeding up the arrival time.

EXAMPLE 2

Consider yourself in a situation where you are a cashier (temporarily) at a department store. There are multiple denominations in the cash register with different quantities of them. Unfortunately, the cash register is broken, and it cannot determine how much cash is in the cash register. Your job is to efficiently figure out how much cash is in the cash register.

A cash register

Some more problems to solve □

- **Problem1:** You want to organize your kitchen pantry, which is cluttered with various items like canned goods, spices, snacks, and baking supplies. Your goal is to find the most efficient way to arrange everything so you can easily find what you need while maximizing the use of space.
- **Problem2:** You need to attend an important meeting at work, but your car has unexpectedly broken down. The meeting is in 45 minutes, and the public transportation option nearest to your home will take around 40 minutes. Your goal is to find the quickest way to get to your office without being late.
- **Problem3:** You want to start a fitness routine to improve your health, but your schedule is quite busy with work and other commitments. You have about 30 minutes of free time each day. Your goal is to create a balanced fitness plan that fits within your available time while helping you achieve your health goals.

PROBLEM ANALYSIS CHART

The Problem Analysis Chart (PAC) is a dependent tool used in troubleshooting problems, especially in programming and systems development. It allows trouble to be broken down into smaller, more manageable parts and is often used to plan and document the steps needed to solve a problem using a programming language like C.

Given Data	Required Results
Section 1: Data given in the problem or provided by the user. These can be known values or general names for data, such as price, quantity, and so forth.	Section 2: Requirements for the output reports. This includes the information needed and the format required.
Processing Required	Solution Alternatives
Section 3: List of processing required. This includes equations or other types of processing, such as sorting, searching, and so forth.	Section 4: List of ideas for the solution of the problem.

An Example of a Problem Analysis Chart (PAC)

In a Problem Analysis Chart there are 4 sections that we need to concentrate on. The Top Left section is the data for the problem we are currently trying to solve, so it will include things such as variables, numerical data, percentile and more. The Top Right section is the desired output we require which is what the problem is based on. The bottom left section is the processing section where the necessary steps are enlisted to ensure the correct output for the required results, and the bottom right section are alternate solutions as a problem can have multiple solutions to achieve the same result.

EXAMPLE

We have a problem in which we want to calculate the gross pay of an employee at the department store that you were previously working as a cashier (assume you are the general manager of that department store). Since you studied well you remember the formula for calculating the formula for gross pay. But I will write it down anyway for ease of use.

GrossPay = Hours*PayRate
Payrate = 20$/HR.
Hours = 12

We need to develop a PAC chart for the above-mentioned problem.

Given Data	Required Results
Hours Pay Rate	Gross Pay
Processing Required	Solution Alternatives
*GrossPay = Hours * PayRate*	1. Define the hours worked and pay rate as constants. *2. Define the hours worked and pay rate as input values.

In the problem listed we identified that Hours and Pay Rate are the data that we are given so it is fed into the top left section of the chart. We require the gross pay of the employee so that goes to our required results section. A formula is required to calculate the gross pay so we can put it in the processing section, finally we can have multiple solutions to the problem as we can have an adaptive pay scale and hour scale.

Some more problems to solve

Create PAC Charts of the following problems

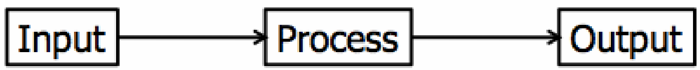
- **Problem1:** Calculate the area and perimeter of a rectangle.
- **Problem2**: Determine the maximum of three numbers.
- **Problem3:** Convert Celsius to Fahrenheit.

INPUT PROCESS OUTPUT

The IPO (Input-Process-Output) model is used in programming and system design for several important reasons. It provides a clear, structured way to conceptualize and develop software by breaking down the process into three fundamental steps: Input, Process, and Output.

IPO Chart shows:

- What data item are input.
- What processing takes place on that data.
- What information will be the result, the output.
- Where in the solution the processing takes place.



EXAMPLE

Input	Processing	Module Reference	Output
Hours Worked Pay Rate	1. Enter Hours Worked 2. Enter Pay Rate 3. Calculate Pay 4. Print Pay 5. End	*Read* *Read* *Calc* *Print* *PayRollControl*	Gross pay

An Example IPO Chart of a Payroll System

In an IPO Chart we have 4 sections. Input, Processing, Module Reference and Output. Like the PAC chart we need to identify what are the input variables in our problem. The processing section has the same effect as the PAC chart but now in a more detailed layout where each step is enlisted. The Module Reference is one change from the PAC chart where each step of the processing section is converted into a command to which a computer will interpret. Such as Read, Calculate, Print, Loop, Write and Program control.

Some more problems to solve

Some more problems to solve

Create IPO Charts of the following problems

- **Problem1:** Calculate the area and perimeter of a rectangle.
- **Problem2**: Determine the maximum of three numbers.
- **Problem3:** Convert Celsius to Fahrenheit.

Find whether a given number is even or odd. Alternatively, the user would take two numbers as input, multiply them and then determine if it is an Odd or Even number (This is the same problem as the above mentioned just create an IPO Chart of it).

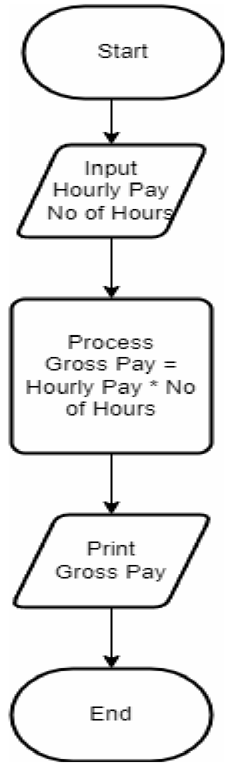
FLOWCHARTS

Flow charts are diagrams showing the exact sequence of logical steps. They use geometrical shapes and arrows to show processes, relationships, and data/process flow. In other words, flowcharts depict decisions and results of them.

In different fields, flowcharts are often used to analyze and manage processes. To put it simply, it helps you visualize what the processes look like. That way, you can see all the bottlenecks and flaws in them. The act of creating a chart is called flowchart.

	**Flowline:** Indicates the flow of logic by connecting symbols.
	**Terminals:** Represents the start and the end of a flowchart.
	**Input/Output:** Used for input & Output operation. 1
	**Processing:** Used for arithmetic operations and data-manipulations.

EXAMPLE



The start and End section of the Flow Chart represent the programs start and end boundaries. The Parallelogram where the Input and Print statements represent the Inputs and Outputs of the Program, and the Square Box represents the Processing required to achieve the desired output.

Some more problems to solve

Create Flow charts of the following problems

- **Problem1:** Calculate the diameter from the radius.
- **Problem2:** Calculate average of three numbers.
- **Problem3:** Calculate the area of rectangle.